FATIMA HUMBATOVA

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EDUCATION

FLORIDA INTERNATIONAL UNIVERSITY Expected Apr 2027
Bachelor of Science in Computer Science | Knight Foundation School of Computing and Information Sciences
Relevant coursework: Data Structures & Algorithms, Software Engineering, Artificial Intelligence, Computer Networks, Digital Forensics, Net-centric Computing

RESEARCH EXPERIENCE & HONORS

AI-Assisted Security Assurance for Defense-Critical Infrastructure April 2026
Research project U.S. Cyber Command Cyber RECon 2026 — Defender Category, First Place

- Investigated AI-assisted assurance cases to assess the completeness and consistency of cybersecurity requirements, identify gaps, and support risk mitigation in critical systems.
- Presented research at the Fifth Annual Cyber Research and Education Conference; team earned first place in the Defender category.
- **Featured interview:** Media.az (June 1, 2026) — national media coverage of Cyber RECon recognition, AI security research, and women in technology.

ExoSearch AI — Exoplanet Detection & Candidate Ranking Mar–Apr 2026
Artificial Intelligence Project, CAI 4002 - Best Project Presentation

- Contributed to a four-person machine-learning project using NASA TESS data, Logistic Regression, and XGBoost to classify and prioritize exoplanet candidates; coordinated team progress and presentation development.
- Independently rebuilt and extended the research pipeline with TIC-disjoint evaluation, model comparisons, reproducibility tests, and an interactive results dashboard.

SOFTWARE ENGINEERING PROJECTS

PhishGuard AI — Explainable Phishing Detection & Adversarial Robustness Oct 2026

- Built a Python/FastAPI and React/TypeScript security research platform for offline phishing URL and email analysis, combining machine-learning predictions with interpretable security findings.
- Evaluated classifiers using domain-disjoint splits, calibration, ablation studies, and controlled URL perturbations, documented dataset bias and model limitations.

TraceTrust — Network Forensics & Threat Detection Sep 2025

- Developed a C++20 network forensics application for PCAP analysis, traffic reconstruction, and explainable detection of suspicious network behavior.
- Implemented automated build and test workflows, structured forensic reports, and a dashboard for investigating network activity.

Digital Evidence Integrity & Timeline Analyzer May 2025

- Developed a Python digital forensics CLI to create SHA-256 evidence snapshots, identify integrity changes, and reconstruct filesystem metadata timelines.
- Designed reproducible investigation workflows with structured outputs, validation tests, and documented forensic limitations.

PROFESSIONAL EXPERIENCE

STARS Peer Tutor — Knight Foundation School of Computing and Information Sciences Aug 2025 – Present

- Tutor 20+ students daily in Data Structures and Algorithms, supporting debugging, programming fundamentals, and independent problem-solving.

Technology Intern, Data Analysis — Konsis Group Mar–Jul 2023

- Developed Python-based data processing pipelines using Pandas and SQL to extract, clean, and analyze financial datasets, identifying spending patterns and supporting data-driven budgeting decisions.
- Prepared structured analytical reports to support financial planning and improve reporting workflows.

Teaching Assistant Intern — İzmir Institute of Technology Jul 2021 – Jan 2022

- Facilitated Calculus I and Physics problem-solving sessions for 15+ students and supported instructional material preparation.

LEADERSHIP & TECHNICAL ENGAGEMENT

INIT FIU — Member & ShellHacks Hackathon Team Lead 2024 - Present

- Led cybersecurity teams at ShellHacks 2025 and 2026, coordinating technical tasks, collaboration, and prototype presentations. Engaged in hackathons, technical workshops, and collaborative learning within FIU's computing community.

Women in Computer Science (WiCS), Florida International University 2023 – Present

- Participated in technical workshops, peer learning, and initiatives supporting women in computing.

TECHNICAL SKILLS & LANGUAGES

Programming & tools: Python, Java, C++, SQL, HTML/CSS, Flask, Pandas, Matplotlib, XGBoost, Git/GitHub, TypeScript, React
Languages: English (fluent), Azerbaijani (native), Russian (native), Turkish (fluent), Spanish (conversational)